Exact Range-Count Envelopes Over Ordered Time Series Under Total-Variation Drift Budgets

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***Abstract*—Threshold-count and range-proportion queries over ordered sensor time series underpin clinical decision support, industrial telemetry, and service-level monitoring, where domain standards map raw measurements into multi-band occupancy summaries such as time in range (TIR) and time in tight range (TITR). In deployed sensing pipelines, however, calibration bias and electrochemical drift perturb observations smoothly over time rather than independently at each sample. Treating per-sample error bounds independently permits implausible step-to-step jumps and produces vacuous count intervals, whereas enforcing a total-variation (TV) coupling budget across consecutive offsets turns a linear threshold scan into a global sequence-optimization problem. Existing dynamic programming (DP) operators evaluate transitions across the full discrete offset domain, incurring substantial state-update overhead when queries probe small or moderate drift budgets across multi-threshold report banks. We propose `LocalWindows`, an exact query operator that computes deterministic minimum and maximum threshold-count envelopes across a bank of TV budgets. First, we establish an extrema-preserving run reduction that contracts fixed always-hit and always-miss samples into scalar constants and merges adjacent identical offset intervals into positive-weight runs without altering any budget-constrained count extremum. Second, we exploit the range-diameter bound of TV-constrained trajectories to cover all feasible offset paths with overlapping $2V$-state local windows, proving that each compressed run is active as a boundary transition in at most four windows ($\leq 4r$ global run-window incidences) while covered runs fold into an $O(1)$ static depth baseline. Third, we introduce a two-pass linear-time state-line transition kernel per window alongside an $O(r+m)$ boundary-prefix fast path for unit drift ($V=1$). Across 6,804 benchmark query cells spanning 211 public continuous glucose monitoring (CGM) series, `LocalWindows` achieves exact bit-for-bit agreement with three exact baselines while reducing prepared-operator runtime at $V \in \{1,4\}$ (median ratios 0.488–0.889, $p \leq 5.7\times10^{-37}$) and reaching up to $10.2\times$ summed-time speedup on 1,152-sample horizons.**

***Index Terms*—time-series data management, uncertain sensor streams, range-count queries, total-variation constraints, interval run compression, dynamic programming**

I. Introduction

Ordered numeric time series are routinely summarized in production databases and clinical analytics engines by counting how many observations fall inside prescribed value intervals and normalizing those counts into proportions [1]–[3]. In continuous glucose monitoring (CGM), international consensus reporting maps a patient's glucose stream $g = (g_1, \ldots, g_n)$ into six standardized threshold bands—including Time in Range (TIR $\in [70, 180]$ mg/dL), Time in Tight Range (TITR $\in [70, 140]$ mg/dL), hypoglycemic bands (TBR70 < 70, TBR54 < 54 mg/dL), and hyperglycemic bands (TAR180 > 180, TAR250 > 250 mg/dL)—which directly govern clinical therapy adjustments and regulatory trial endpoints [1], [4], [5]. Analogous threshold-occupancy queries arise in Intensive Care Unit (ICU) physiological alarms, battery management telemetry, and environmental compliance auditing.

Despite the precision of digital reporting pipelines, physical sensors exhibit systematic calibration offset and gradual time-varying drift [1], [2]. If each sample g_i is perturbed by an unknown integer offset $\delta_i \in D = \{-R, \ldots, R\}$, evaluating a threshold predicate $g_i + \delta_i \in [L, U]$ becomes an interval-uncertainty query: what are the exact minimum and maximum possible hit counts $[\mathrm{lo}(V), \mathrm{hi}(V)]$ over all feasible offset trajectories $\delta = (\delta_1, \ldots, \delta_n)$? Answering this question accurately exposes a fundamental tension between physical realism and query evaluation complexity:

- **Vacuousness of Uncoupled Row Bounds:** Standard uncertain-database range queries treat per-sample uncertainty intervals independently [6]–[9], implicitly allowing δ_i to jump from $+R$ to $-R$ between consecutive minutes. On public record `BIG001` (Fig. 1a,c), uncoupled row evaluation reports a TITR count interval of $[33, 96]$ (34.4%–100.0%, width 63), which is too wide to support clinical interpretation.
- **Computational Bottleneck of Coupled Drift DP:** Because physical sensor drift evolves gradually, consecutive offsets satisfy a total-variation (TV) budget $\mathrm{TV}(\delta) = \sum_{i=2}^{n} |\delta_i - \delta_{i-1}| \leq V$. Under $\mathrm{TV}(\delta) \leq V$, every sample's offset is coupled to its neighbors [10]–[12]. As illustrated on `BIG001` at $V = 4$ (Fig. 1a,c), temporal coupling narrows the TITR envelope by 54.0% to $[63, 92]$ (65.6%–95.8%, width 29): to maximize hits, the optimal trajectory starts at $\delta_1^{\max} = +11$ to lift nine early hypoglycemic readings ($i = 1 \ldots 10$) into $[70, 140]$ and then expends exactly 4 variation units ($+11 \to +10 \to +7$) before the post-meal peak ($i = 16$) so $g_{16}+7 = 140$ stays inside the upper threshold. However, computing exact count envelopes over a bank of budgets $\mathcal{V} = \{v_1 < \cdots < v_K\}$ via sample-level dynamic programming scales as $O(nV_{\max}m^2)$ over $m = |D| = 2R+1$ states, creating a severe bottleneck when evaluating multi-predicate report banks across large



Fig. 1. Phenomenon, algorithmic localization, and empirical verification on public CGM record `BIG001` ($n = 96$, TITR $= [70, 140]$ mg/dL, offset domain $D = \{-30, \ldots, 30\}$) and the 211-series cohort. (a) Under a coupled variation budget $V = 4$, the maximum-attaining trajectory $g + \delta^{\max}$ ($\delta^{\max}$: $+11 \to +10 \to +7$, TV $= 4$) rescues 9 of 10 early low readings ($i = 1 \ldots 10$) into $[70, 140]$ and drifts down to $+7$ before the post-meal peak ($i = 16$, $g_{16} = 133$) so $g_{16} + \delta_{16} = 140$ remains in range, reaching $92/96$ (95.8%) vs. $84/96$ (87.5%) nominal and $63/96$ (65.6%) minimum. (b) In state space, $33/96$ always-hit rows ($I_i = [-30, 30]$, gray) contract into scalar $c_1 = 33$, active rows merge into $r = 56$ runs, and `LocalWindows` confines the $V = 4$ DP to overlapping $2V$-state windows such as $W_j = [+4, +12]$ (width $9 \ll 61$), evaluating only boundary endpoints crossing W_j ($\leq 4r$ incidences, pruning 67.1% of DP updates). (c) Temporal coupling narrows the unconstrained independent-row interval $[33, 96]$ (width 63) by 54.0% to $[63, 92]$ (width 29) at $V = 4$ and saturates at $[63, 95]$ for $V \geq 16$. (d–f) Inner-loop DP work drops to 0% at $V = 1$ and 32.9% at $V = 4$, yielding $3.64\times$ and $1.94\times$ query speedups on `BIG001` and $3.03\times$ and $1.72\times$ cohort median speedups across 211 series ($p < 10^{-36}$).

cohorts.

A natural question is how to exploit both temporal redundancy along the time axis and excursion locality along the state axis without sacrificing exactness. When each sample g_i is compiled against a threshold $[L, U]$ into a feasible integer offset interval $I_i = [\ell_i, u_i] \subseteq D$, two structural properties emerge (Fig. 1b): (1) many samples lie deep inside or outside the threshold band, yielding fixed always-hit rows ($I_i = D$) or always-miss rows ($I_i = \emptyset$) whose indicator values are invariant to δ_i, while adjacent active samples frequently compile to identical intervals; and (2) under a small or moderate variation budget V, any trajectory δ satisfying $\mathrm{TV}(\delta) \leq V$ has range diameter $\max_t \delta_t - \min_t \delta_t \leq V$, meaning the entire trajectory is confined to a contiguous sub-interval of D of width at most $V + 1 \ll m$.

Existing sequence-DP optimizations fail to exploit these two properties jointly. Run-length compression alone reduces the time horizon from n samples to r weighted runs, and distance-transform linear sweeps reduce per-step transitions from $O(m^2)$ to $O(m)$ [13], but global-state DP still updates all m offset states across every run and budget [12], [14]. Conversely, partitioning the state space around interval endpoints without overlapping excursion covers breaks trajectories that drift across block boundaries.

In this paper, we present `LocalWindows`, an exact query operator for multi-budget threshold-count envelopes over TV-coupled time series. Figure 2 summarizes our three-stage architecture. First, we prove an *extrema-preserving run compression theorem* showing that fixed rows ($I_i = D$ and $I_i = \emptyset$) can be contracted out of the sequence and adjacent duplicate intervals merged into r positive-weight runs (I_t, w_t) without altering any TV-constrained minimum or maximum count. Second, for any budget V, we tile the offset grid D with *overlapping* $2V$*-state windows* $W_j = B_j \cup B_{j+1}$ of width $|W_j| \leq 2V + 1$, guaranteeing that every feasible trajectory lies entirely inside at least one window W_j. Within each window W_j, runs disjoint from W_j contribute 0, runs covering W_j contribute a constant weight $+w_t$ recovered in $O(1)$ time from a static event sweep $d(z)$, and only runs whose endpoints cross W_j are nominated as active boundary runs R_j. Because each run endpoint falls in at most two adjacent windows, the total number of active run-window incidences globally is bounded by $\sum_j |R_j| \leq 4r$. Third, we design a two-pass $O(|W_j|)$ state-line transition kernel per active window and a specialized $O(r+m)$ boundary-prefix fast path for unit drift ($V = 1$) that eliminates DP tables entirely.

Our main contributions are summarized as follows:

- **Extrema-Preserving Sample-to-Run Reduction (Section II):** We formulate exact TV-coupled range-count envelopes over ordered time series and prove (Lemma 1 and Theorem 1) that fixed-row contraction and adjacent equal-interval run merging preserve exact minimum and maximum counts for every budget $V \geq 0$.
- **Overlapping $2V$-Window Decomposition & $\leq 4r$ Incidence Bound (Section III):** We construct an overlapping $2V$-state window cover (Lemma 2) that localizes every feasible trajectory, coupled with boundary-edge run nomination that bounds active run-window incidences to

TABLE I
CONSENSUS CLINICAL GLYCEMIC THRESHOLD PREDICATES AND THEIR COMPILED INTEGER OFFSET INTERVALS $I_i \subseteq D = \{-R, \ldots, R\}$ FOR AN INTEGER READING g_i (ALL THRESHOLDS IN MG/DL).

Metric	Band	Clinical Meaning	Compiled Interval $I_i \subseteq D$
TIR	$[70, 180]$	In-range	$[\max(-R, 70 - g_i),\ \min(R, 180 - g_i)]$
TITR	$[70, 140]$	Tight in-range	$[\max(-R, 70 - g_i),\ \min(R, 140 - g_i)]$
TBR70	$(-\infty, 69]$	L1 hypoglyc.	$[-R,\ \min(R, 69 - g_i)]$
TBR54	$(-\infty, 53]$	L2 hypoglyc.	$[-R,\ \min(R, 53 - g_i)]$
TAR180	$[181, \infty)$	L1 hyperglyc.	$[\max(-R, 181 - g_i),\ R]$
TAR250	$[251, \infty)$	L2 hyperglyc.	$[\max(-R, 251 - g_i),\ R]$

$\sum_j |R_j| \le 4r$ (Lemma 3), $O(1)$ anchor-state baseline restoration, linear-time two-pass transition sweeps, and an $O(r+m)$ boundary-prefix solver for $V = 1$ (Proposition 2 and Theorem 2).

- **Regime Analysis & Exact Multi-Baseline Verification (Section IV):** We evaluate `LocalWindows` against three exact DP baselines (`SparseGlobal`, `GapComponents`, and `RunTailFullG`) across 211 public CGM series (6,804 query cells per mode). All four operators achieve 100% bit-for-bit output agreement across every cell and budget. `LocalWindows` significantly accelerates prepared queries in the localized drift regime $V \in \{1, 4\}$ ($3.03\times$ and $1.72\times$ cohort speedups, $p \le 5.7 \times 10^{-37}$), reaches $10.2\times$ speedup on $n = 1{,}152$ horizons, and cleanly characterizes the structural crossover at $V \ge 16$ where window covers approach the full state domain.

II. PROBLEM FORMULATION AND EXACT RUN REDUCTION

A. Time-Series Threshold Queries Under Coupled Drift

Let $g = (g_1, \ldots, g_n)$ be an ordered sequence of $n \ge 1$ sensor observations sampled at regular timestamps. A range-count predicate $P = [L, U]$ (which may be two-sided, lower-unbounded $(-\infty, U]$, or upper-unbounded $[L, \infty)$, as cataloged in Table I) tests whether each perturbed reading $g_i + \delta_i$ falls inside $[L, U]$. Here $\delta = (\delta_1, \ldots, \delta_n) \in D^n$ denotes an integer offset trajectory taking values in the discrete offset domain

$$D = \{-R, -R+1, \ldots, R\}, \qquad m = |D| = 2R + 1, \quad (1)$$

where $R \in \mathbb{Z}_{\ge 0}$ specifies the maximum per-sample calibration error radius (e.g., $R \in \{15, 30, 120\}$ mg/dL).

To capture the physical constraint that sensor drift evolves gradually over time rather than jumping arbitrarily between adjacent readings, we constrain the discrete total variation of the offset trajectory δ:

$$\mathrm{TV}(\delta) = \sum_{i=2}^{n} |\delta_i - \delta_{i-1}|, \qquad \mathcal{F}(V) = \big\{\delta \in D^n : \mathrm{TV}(\delta) \le V\big\}, \quad (2)$$

where $V \in \mathbb{Z}_{\ge 0}$ is a user-specified variation budget. When $V = 0$, $\mathcal{F}(0)$ restricts δ to a constant calibration shift $\delta_1 = \cdots = \delta_n = z \in D$. As V increases, $\mathcal{F}(V)$ permits up to V cumulative integer steps of drift across the horizon. Conversely, when $V \ge 2R(n-1)$, the variation constraint becomes non-binding and $\mathcal{F}(V) = D^n$ reduces to unconstrained independent per-sample uncertainty.

For any fixed reading g_i and threshold band $[L, U]$, the set of integer offsets $\delta_i \in D$ that satisfy the predicate $g_i + \delta_i \in [L, U]$ forms a contiguous integer interval $I_i \subseteq D$:

$$I_i = \big\{\delta_i \in D : L \le g_i + \delta_i \le U\big\} = [\ell_i, u_i] \cap D, \quad (3)$$

where $\ell_i = \max(-R, \lceil L - g_i \rceil)$ and $u_i = \min(R, \lfloor U - g_i \rfloor)$, with $I_i = \emptyset$ whenever $\ell_i > u_i$. Using the indicator $\mathbf{1}\{\delta_i \in I_i\} \in \{0, 1\}$, the threshold hit count achieved by trajectory $\delta \in \mathcal{F}(V)$ is

$$C(\delta) = \sum_{i=1}^{n} \mathbf{1}\{\delta_i \in I_i\}. \quad (4)$$

Definition 1 (Multi-Budget Range-Count Envelope Query)**.** *Given an ordered time series $g = (g_1, \ldots, g_n)$, a threshold predicate $P = [L, U]$, an offset domain $D = \{-R, \ldots, R\}$, and an ordered bank of variation budgets $\mathcal{V} = \{v_1 < v_2 < \cdots < v_K\} \subset \mathbb{Z}_{\ge 0}$, compute for every $v \in \mathcal{V}$ the exact minimum and maximum attainable hit counts*

$$\mathrm{lo}(v) = \min_{\delta \in \mathcal{F}(v)} C(\delta), \qquad \mathrm{hi}(v) = \max_{\delta \in \mathcal{F}(v)} C(\delta), \quad (5)$$

together with the corresponding exact range-proportion interval $\big[100 \cdot \mathrm{lo}(v)/n,\ 100 \cdot \mathrm{hi}(v)/n\big]$.

Because $\mathcal{F}(v) \subseteq \mathcal{F}(v')$ whenever $v \le v'$, the exact envelope widens monotonically with the variation budget: $\mathrm{lo}(v') \le \mathrm{lo}(v) \le C(\mathbf{0}) \le \mathrm{hi}(v) \le \mathrm{hi}(v')$. Moreover, because the maximum step between consecutive samples is $|\delta_i - \delta_{i-1}| \le 2R$, any budget $v \ge V_{\mathrm{sat}} = 2R(n-1)$ saturates at the independent per-sample bounds $\mathrm{lo}_{\mathrm{ind}} = \sum_{i=1}^{n} \mathbf{1}\{I_i = D\}$ and $\mathrm{hi}_{\mathrm{ind}} = \sum_{i=1}^{n} \mathbf{1}\{I_i \ne \emptyset\}$.

B. Extrema-Preserving Sample-to-Run Reduction

Evaluating (5) directly on the n raw intervals $(I_1, \ldots, I_n)$ by dynamic programming requires tracking state transitions at every sample $i \in \{1, \ldots, n\}$. However, compiled interval sequences exhibit two forms of structural redundancy that can be eliminated exactly before any dynamic programming takes place.

First, consider a sample i whose compiled interval is either the full domain ($I_i = D$, an *always-hit* row) or empty ($I_i = \emptyset$, an *always-miss* row). For such a sample, the indicator $\mathbf{1}\{\delta_i \in I_i\}$ equals 1 (if $I_i = D$) or 0 (if $I_i = \emptyset$) regardless of the chosen offset $\delta_i \in D$.

Lemma 1 (Fixed-Row Contraction)**.** *Let $c_1 = |\{i : I_i = D\}|$ and $c_0 = |\{i : I_i = \emptyset\}|$ denote the counts of always-hit and always-miss samples in $(I_1, \ldots, I_n)$, and let $(I_{i_1}, I_{i_2}, \ldots, I_{i_{\tilde{n}}})$ be the subsequence of remaining active samples where $\emptyset \subsetneq I_{i_k} \subsetneq D$. For every variation budget $V \ge 0$:*

- *If $\tilde{n} = 0$ (all rows are fixed), then $\mathrm{lo}(V) = \mathrm{hi}(V) = c_1$.*
- *If $\tilde{n} \ge 1$, then for both minimization and maximization, the exact count extrema over $(I_1, \ldots, I_n)$ equal c_1 plus the exact count extrema over the contracted sequence $(I_{i_1}, \ldots, I_{i_{\tilde{n}}})$ under the same budget V.*

Proof. For any trajectory $\delta = (\delta_1, \ldots, \delta_n) \in \mathcal{F}(V)$, the objective (4) decomposes into $C(\delta) = c_1 + \sum_{k=1}^{\tilde{n}} \mathbf{1}\{\delta_{i_k} \in I_{i_k}\}$. By the triangle inequality on $\mathbb{Z}$, restricting δ to the active indices $\tilde{\delta} = (\delta_{i_1}, \ldots, \delta_{i_{\tilde{n}}})$ satisfies

$$\mathrm{TV}(\tilde{\delta}) = \sum_{k=2}^{\tilde{n}} |\delta_{i_k} - \delta_{i_{k-1}}| \le \sum_{i=2}^{n} |\delta_i - \delta_{i-1}| = \mathrm{TV}(\delta) \le V.$$

Conversely, any feasible active trajectory $\tilde{\delta} \in D^{\tilde{n}}$ with $\mathrm{TV}(\tilde{\delta}) \le V$ can be extended to a full trajectory $\delta \in D^n$ with identical total variation $\mathrm{TV}(\delta) = \mathrm{TV}(\tilde{\delta}) \le V$ and identical objective value by setting $\delta_i = \tilde{\delta}_1$ for $i < i_1$, $\delta_i = \tilde{\delta}_{\tilde{n}}$ for $i > i_{\tilde{n}}$, and $\delta_i = \tilde{\delta}_k$ for all fixed indices $i \in (i_k, i_{k+1})$. Thus the set of attainable objective values is identical in both directions. □

Second, after removing all fixed rows, suppose the contracted sequence contains a contiguous block of identical active intervals $I_a = I_{a+1} = \cdots = I_b = I$.

Theorem 1 (Extrema-Preserving Duplicate Run Merging)**.** *Let $(I_1, w_1), (I_2, w_2), \ldots, (I_r, w_r)$ be the run-length compressed representation obtained after fixed-row contraction by merging each maximal contiguous block of $w_t = b - a + 1 \ge 1$ identical active intervals into a single weighted run (I_t, w_t) with $\emptyset \subsetneq I_t \subsetneq D$ and $I_t \ne I_{t-1}$ for all $t \in \{2, \ldots, r\}$. Then for every budget $V \ge 0$:*

$$\begin{aligned} \mathrm{lo}(V) &= c_1 + \min_{\eta \in D^r : \mathrm{TV}(\eta) \le V} \sum_{t=1}^{r} w_t \, \mathbf{1}\{\eta_t \in I_t\}, \\ \mathrm{hi}(V) &= c_1 + \max_{\eta \in D^r : \mathrm{TV}(\eta) \le V} \sum_{t=1}^{r} w_t \, \mathbf{1}\{\eta_t \in I_t\}. \end{aligned} \tag{6}$$

Proof. Any run-level trajectory $\eta = (\eta_1, \ldots, \eta_r) \in D^r$ with $\mathrm{TV}(\eta) \le V$ expands to a sample-level trajectory $\tilde{\delta}$ by repeating η_t across the w_t identical rows of block t. Because constant repetition incurs zero internal variation ($|\eta_t - \eta_t| = 0$), $\mathrm{TV}(\tilde{\delta}) = \mathrm{TV}(\eta) \le V$ and its hit count is $\sum_{t=1}^{r} w_t \mathbf{1}\{\eta_t \in I_t\}$. Conversely, let $\tilde{\delta}$ be any feasible trajectory on the unmerged active rows with $\mathrm{TV}(\tilde{\delta}) \le V$. Within block $t = \{a, \ldots, b\}$ where all rows share the same interval I_t, let $s_{\max} = \arg\max_{s \in \{a, \ldots, b\}} \mathbf{1}\{\tilde{\delta}_s \in I_t\}$ (for maximization) or $s_{\min} = \arg\min_{s \in \{a, \ldots, b\}} \mathbf{1}\{\tilde{\delta}_s \in I_t\}$ (for minimization). Replacing $\tilde{\delta}_s$ for every $s \in \{a, \ldots, b\}$ by the single representative state $\eta_t = \tilde{\delta}_{s_{\max}}$ (or $\tilde{\delta}_{s_{\min}}$) can only increase (or decrease) the block's hit count to $w_t \mathbf{1}\{\eta_t \in I_t\}$, while by the triangle inequality

$$|\eta_t - \eta_{t-1}| = |\tilde{\delta}_{s_t} - \tilde{\delta}_{s_{t-1}}| \le \sum_{s = s_{t-1}+1}^{s_t} |\tilde{\delta}_s - \tilde{\delta}_{s-1}|,$$

so summing over $t = 2, \ldots, r$ yields $\mathrm{TV}(\eta) \le \mathrm{TV}(\tilde{\delta}) \le V$. Hence optimal run-level and sample-level extrema coincide. □

Worked Trace Contraction: Consider a 6-sample slice on $D = \{-3, \ldots, 3\}$ (Fig. 2, Stage 1 bottom) with compiled intervals $I_1 = [-2, 0]$, $I_2 = [-2, 0]$, $I_3 = [-3, 3] = D$, $I_4 = [0, 2]$, $I_5 = \emptyset$, and $I_6 = [2, 3]$. Lemma 1 contracts $I_3 = D$ into $c_1 = 1$ and $I_5 = \emptyset$ into $c_0 = 1$; Theorem 1 merges $I_1 = I_2 = [-2, 0]$ into $(I_1 = [-2, 0], w_1 = 2)$, followed by $(I_2 = [0, 2], w_2 = 1)$ and $(I_3 = [2, 3], w_3 = 1)$, reducing $n = 6$ samples to $r = 3$ compressed runs. Across our 211-series CGM benchmark, this reduction contracts $n = 96$ observations to a median of $r = 24$ runs.

III. Localized Excursion Decomposition and Operator Design

Even on the compressed run sequence $\{(I_t, w_t)\}_{t=1}^{r}$, standard dynamic programming over the full state domain D maintains tables of size $(V + 1) \times m$ and performs $O(rVm)$ state updates. We now show how `LocalWindows` eliminates global state-space scans for small and moderate variation budgets by decomposing the offset domain into overlapping local windows and nominating only boundary-crossing runs.

A. Static Event Sweep for $V = 0$ and Depth Lookup Bank

Before evaluating positive variation budgets $V > 0$, we compute for every offset $z \in D$ the exact static hit count $d(z)$ achieved by a constant trajectory $\eta = (z, \ldots, z)$:

$$d(z) = c_1 + \sum_{t=1}^{r} w_t \, \mathbf{1}\{z \in I_t\}, \qquad z \in D. \tag{7}$$

Rather than evaluating (7) in $O(rm)$ time, we allocate a difference array $\Delta[-R \ldots R + 1]$ initialized to zero. Each run $I_t = [\ell_t, u_t]$ adds $+w_t$ at index ℓ_t and $-w_t$ at index $u_t + 1$ in $O(1)$ time. A single prefix sum over $z = -R, \ldots, R$ reconstructs all m values of $d(z)$ in $O(r+m)$ time, immediately yielding the exact $V = 0$ envelope

$$\alpha = \mathrm{lo}(0) = \min_{z \in D} d(z), \qquad \beta = \mathrm{hi}(0) = \max_{z \in D} d(z). \tag{8}$$

Crucially, the static depth array $d(z)$ is retained in memory as an $O(1)$ lookup bank to recover covered-run baselines in Stage 2 without scanning covered runs.

B. Overlapping $2V$-State Window Cover

Consider any variation budget $V \ge 1$. A fundamental geometric property of total variation is that the range diameter of any trajectory is bounded by its total variation:

$$\mathrm{TV}(\eta) = \sum_{t=2}^{r} |\eta_t - \eta_{t-1}| \le V \implies \max_{1 \le t \le r} \eta_t - \min_{1 \le t \le r} \eta_t \le V. \tag{9}$$

Consequently, no feasible trajectory under budget V can visit states more than V units apart. We exploit (9) to cover all feasible trajectories using overlapping local windows of width at most $2V + 1$.

Lemma 2 (Overlapping $2V$-Window Cover)**.** *For a fixed budget $V \ge 1$, partition the ordered domain $D = \{-R, \ldots, R\}$ into $B = \lceil m/(V+1) \rceil$ contiguous blocks $B_0, B_1, \ldots, B_{B-1}$, each of size at most $b = V + 1$. If $B = 1$, define a single window $W_0 = D$. If $B \ge 2$, define $J = B - 1$ overlapping windows by taking the union of each pair of adjacent blocks:*

$$W_j = B_j \cup B_{j+1} = [a_j, b_j] \subseteq D, \qquad j \in \{0, 1, \ldots, B - 2\}, \tag{10}$$

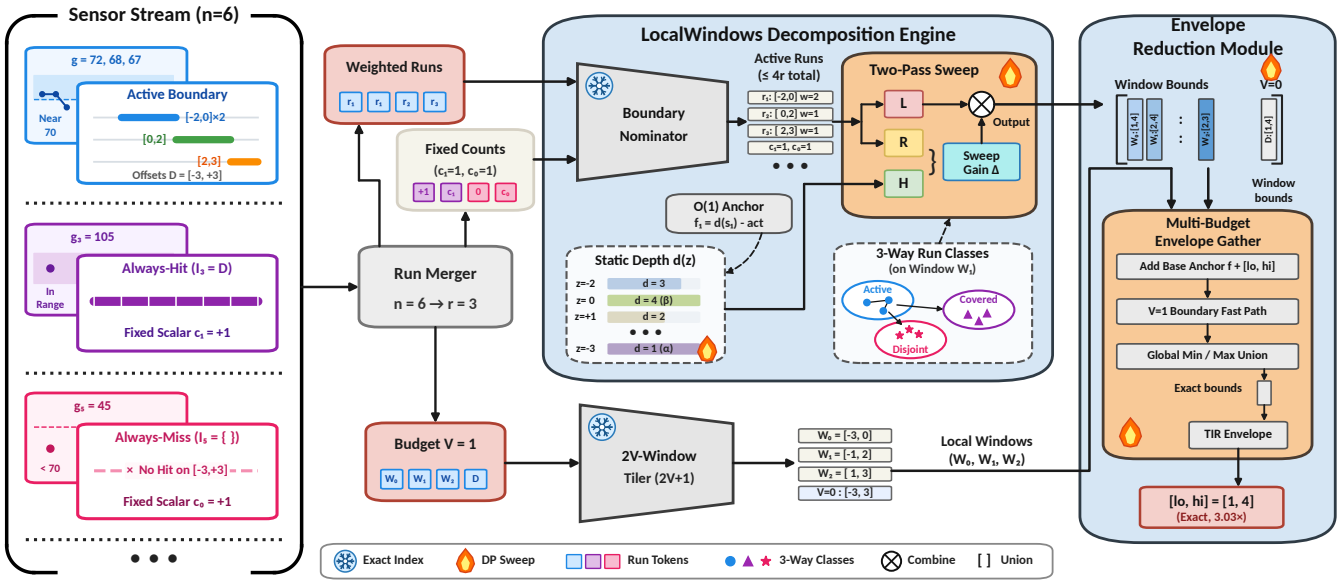


Fig. 2. System architecture and mathematical pipeline of `LocalWindows`. **Stage 1 (Left):** Raw time-series observations g_i and threshold band $[L, U]$ compile into integer offset intervals $I_i \subseteq D$; fixed always-hit ($I_i = D$) and always-miss ($I_i = \emptyset$) rows contract into scalars c_1, c_0 (Lemma 1), and adjacent identical intervals merge into r weighted runs (I_t, w_t) (Theorem 1). **Static Sweep & $2V$-Window Cover (Middle-Left):** A single $O(r+m)$ difference-array sweep computes static depths $d(z)$, yielding exact $V=0$ bounds $[\alpha, \beta]$ and feeding anchor baselines into the overlapping $2V$-state window cover $W_j = B_j \cup B_{j+1}$ ($|W_j| \le 2V+1$, Lemma 2). **Stage 2 (Center):** Within each window W_j, compressed runs classify into disjoint misses (0), covered hits ($+w_t$), or active boundary runs R_j ($\sum_j |R_j| \le 4r$, Lemma 3); covered weights are recovered in $O(1)$ per active window via $f_j = d(s_j) - \sum_{t \in R_j} w_t \mathbf{1}\{s_j \in I_t\}$, and active runs are processed by a two-pass $O(|W_j|)$ state-line sweep ($V \ge 2$) or an $O(r+m)$ boundary-prefix scan ($V=1$). **Stage 3 (Right):** Active window optima combine with $[\alpha, \beta]$ under local saturation caps B_j to produce exact multi-budget count and percentage envelopes $[\mathrm{lo}(v), \mathrm{hi}(v)]$ (Theorem 2).

where $|W_j| \le 2V+2$ (and every span of $V+1$ consecutive states lies within a sub-window of width $\le 2V+1$). Then for every trajectory $\eta \in D^r$ satisfying $\mathrm{TV}(\eta) \le V$, there exists at least one window index j such that $\eta_t \in W_j$ for all $t \in \{1, \dots, r\}$.

Proof. Let $\underline{\eta} = \min_t \eta_t$ and $\overline{\eta} = \max_t \eta_t$. By (9), $\overline{\eta} - \underline{\eta} \le V < b$. Let B_j be the block containing $\underline{\eta}$. Because each block (except possibly the last) has width $b = V+1$, the upper extremum $\overline{\eta} \le \underline{\eta} + V$ must lie either in the same block B_j or in the immediately succeeding block B_{j+1}. In either case, $[\underline{\eta}, \overline{\eta}] \subseteq B_j \cup B_{j+1} = W_j$ (with $j = \max(0, B-2)$ if $\underline{\eta} \in B_{B-1}$). Thus $\eta \in W_j^r$. □

Lemma 2 implies that global optimization over $\mathcal{F}(V)$ decomposes exactly into independent local optimizations over trajectories confined to each window W_j:

$$\mathrm{lo}(V) = \min_j \min_{\eta \in W_j^r : \mathrm{TV}(\eta) \le V} \left(c_1 + \sum_{t=1}^{r} w_t \, \mathbf{1}\{\eta_t \in I_t\} \right),$$
$$\mathrm{hi}(V) = \max_j \max_{\eta \in W_j^r : \mathrm{TV}(\eta) \le V} \left(c_1 + \sum_{t=1}^{r} w_t \, \mathbf{1}\{\eta_t \in I_t\} \right). \tag{11}$$

C. Three-Way Run Classification and $\le 4r$ Incidence Bound

If every window W_j had to iterate through all r runs, decomposing D into J windows would still cost $O(Jr)$ run inspections. Our second key insight is that relative to a local window $W_j = [a_j, b_j]$, almost all runs are either completely disjoint from W_j or completely cover W_j (Fig. 2, Stage 2 top):

1) **Disjoint Miss ($I_t \cap W_j = \emptyset$):** For every state $z \in W_j$, $\mathbf{1}\{z \in I_t\} = 0$. Run t contributes zero to every trajectory in W_j^r and is pruned.
2) **Covered Hit ($W_j \subseteq I_t$, i.e., $\ell_t \le a_j$ and $b_j \le u_t$):** For every state $z \in W_j$, $\mathbf{1}\{z \in I_t\} = 1$. Run t contributes a constant weight $+w_t$ to every trajectory in W_j^r and requires no dynamic programming transitions.
3) **Active Boundary Run ($t \in R_j$):** Run I_t partially intersects W_j ($\emptyset \subsetneq I_t \cap W_j \subsetneq W_j$). This occurs if and only if at least one of its two discrete boundary transition points—its left entry point ℓ_t or its right exit point $u_t + 1$—falls strictly inside $[a_j + 1, b_j]$ (or $\ell_t = a_j$ with $u_t < b_j$).

Lemma 3 (Global Boundary-Incidence Sparsity Bound)**.** *Let $R_j = \{t \in \{1, \dots, r\} : \emptyset \subsetneq I_t \cap W_j \subsetneq W_j\}$ denote the ordered list of active boundary runs for window W_j. Across all windows $W_0, \dots, W_{J-1}$:*

$$\sum_{j=0}^{J-1} |R_j| \le 4r. \tag{12}$$

Moreover, the active lists $\{R_j\}_{j=0}^{J-1}$ can be constructed in $O(r+J)$ total time without testing all (t, j) pairs.

Proof. A run $I_t = [\ell_t, u_t]$ belongs to R_j if and only if W_j contains at least one of the two boundary transitions $\ell_t \in$

$[a_j+1, b_j]$ or $u_t+1 \in [a_j+1, b_j]$. Because the underlying blocks $B_0, \ldots, B_{B-1}$ partition D disjointly and each window $W_j = B_j \cup B_{j+1}$ covers exactly two consecutive blocks, any integer coordinate $x \in D$ belongs to at most two windows (specifically W_{k-1} and W_k, where $x \in B_k$). Therefore, ℓ_t nominates run t to at most two windows, and u_t+1 nominates run t to at most two windows. Summing over all r runs yields $\sum_j |R_j| \le 2r + 2r = 4r$. Computing the block index $k = \lfloor (x+R)/(V+1) \rfloor$ for $x \in \{\ell_t, u_t+1\}$ in $O(1)$ arithmetic time nominates run t directly to its ≤ 4 candidate windows in $O(r)$ total time. □

$O(1)$ **Static Baseline Restoration:** For each window W_j, let $C_j = \{t : W_j \subseteq I_t\}$ be the set of covered runs and let $f_j = c_1 + \sum_{t \in C_j} w_t$ denote the constant baseline score contributed by always-hit rows and covered runs inside W_j. Instead of enumerating C_j, we evaluate the precomputed static depth $d(s_j)$ at the window's left anchor state $s_j = a_j = \min W_j$. By definition, a run covers s_j if and only if it either covers all of W_j ($t \in C_j$) or belongs to R_j and happens to contain s_j. Subtracting the active boundary runs that contain s_j recovers f_j in $O(|R_j|)$ time ($O(r)$ globally across all windows):

$$f_j = d(s_j) - \sum_{t \in R_j} w_t \, \mathbf{1}\{s_j \in I_t\}. \tag{13}$$

Furthermore, if a window has no active boundary runs ($R_j = \emptyset$), every trajectory confined to W_j achieves the constant score $f_j = d(s_j)$, which is already accounted for by the static $V = 0$ extrema $[\alpha, \beta] = [\min_z d(z), \max_z d(z)]$. Thus `LocalWindows` completely skips inactive windows and evaluates only the active window set $\mathcal{A} = \{j : R_j \neq \emptyset\}$, whose cardinality satisfies $|\mathcal{A}| \le \min(J, 4r)$.

D. Linear-Time State-Line Sweeps and $V = 1$ Fast Path

Two-Pass $O(|W_j|)$ Transition Kernel ($V \ge 2$): Fix an active window $W_j = \{e_0, e_1, \ldots, e_{m_j-1}\}$ of width $m_j = |W_j| \le 2V+1$ and its active run list $R_j = (t_1, \ldots, t_{N_j})$ where $N_j = |R_j|$. Because runs outside R_j contribute constant weight f_j across all states in W_j, an optimal trajectory needs only change state between active runs in R_j. Moreover, because the maximum state step inside W_j is $m_j - 1$, a trajectory across N_j active runs can usefully expend at most

$$B_j = \min\big(V_{\max},\, \max(0, N_j - 1)(m_j - 1)\big) \tag{14}$$

units of variation; for any requested budget $v > B_j$, the local optima saturate at $\mathrm{lo}_j(v) = \mathrm{lo}_j(B_j)$ and $\mathrm{hi}_j(v) = \mathrm{hi}_j(B_j)$.

Let $H_v(q)$ and $L_v(q)$ denote the maximum and minimum active-run hit counts up to the current active run, ending at local state index $q \in \{0, \ldots, m_j - 1\}$ with cumulative variation $v \in \{0, \ldots, B_j\}$. After initializing $H_0(q) = L_0(q) = w_{t_1} \mathbf{1}\{e_q \in I_{t_1}\}$ and $H_v(q) = -\infty, L_v(q) = +\infty$ for $v \ge 1$, advancing to the next active run $t \in R_j$ requires computing

$$\begin{aligned} Q_v(q) &= \max_{0 \le p < m_j,\, |q-p| \le v} H_{v-|q-p|}(p), \\ H_v^{\mathrm{new}}(q) &= w_t \mathbf{1}\{e_q \in I_t\} + Q_v(q). \end{aligned} \tag{15}$$

Evaluating (15) naively over all (q, p) pairs takes $O(m_j^2)$ time per budget. Instead, we decompose $|q-p|$ into leftward steps ($p \le q$) and rightward steps ($p \ge q$) by maintaining two running envelopes across budgets:

$$\begin{aligned} A_v(q) &= \max\big\{H_v(q),\, A_{v-1}(q-1)\big\}, \quad q = 0, \ldots, m_j - 1, \\ Z_v(q) &= \max\big\{H_v(q),\, Z_{v-1}(q+1)\big\}, \quad q = m_j - 1, \ldots, 0, \end{aligned} \tag{16}$$

with out-of-range neighbors set to $-\infty$. Unrolling (16) gives $A_v(q) = \max_{p \le q, q-p \le v} H_{v-(q-p)}(p)$ and $Z_v(q) = \max_{p \ge q, p-q \le v} H_{v-(p-q)}(p)$, so $Q_v(q) = \max\{A_v(q), Z_v(q)\}$ is obtained in $O(1)$ operations per state and $O(m_j)$ per budget v. The minimum table $L_v(q)$ updates symmetrically with $\min$ replacing $\max$.

Proposition 1 (Local Budget Saturation and Bank Table Reuse)**.** *For any active window W_j of width $m_j = |W_j|$ with $N_j = |R_j|$ active boundary runs, any trajectory $\eta \in W_j^r$ exceeding variation $B_j = \min(V_{\max}, \max(0, N_j - 1)(m_j - 1))$ can be replaced by $\tilde{\eta} \in W_j^r$ with $\mathrm{TV}(\tilde{\eta}) \le B_j$ achieving identical hit count. Thus, for an ordered budget bank $\mathcal{V} = \{v_1 < \cdots < v_K\}$ ($V_{\max} = v_K$), a single sweep to depth B_j yields exact window extrema $\mathrm{lo}_j(v_k) = \mathrm{lo}_j(\min(v_k, B_j))$ and $\mathrm{hi}_j(v_k) = \mathrm{hi}_j(\min(v_k, B_j))$ for all $v_k \in \mathcal{V}$ ($v_k \ge 2$).*

Proof. Every run $t \notin R_j$ is either disjoint from or covers W_j, so $\mathbf{1}\{z \in I_t\}$ is constant over $z \in W_j$. Holding $\tilde{\eta}_t$ constant outside R_j incurs zero variation while preserving all hit counts, leaving $\max(0, N_j - 1)$ transitions of step size $\le m_j - 1$. Sequential evaluation of (16) over $v = 0, \ldots, B_j$ populates all $v_k \le V_{\max}$ in one pass. □

Boundary-Prefix Fast Path for Unit Drift ($V = 1$): When $V = 1$, any feasible trajectory $\eta \in D^r$ with $\mathrm{TV}(\eta) \le 1$ is either constant ($\mathrm{TV}(\eta) = 0$, already solved by $[\alpha, \beta]$) or makes a single unit step from z to $z + \sigma$ ($\sigma \in \{-1, +1\}$) between some split run $t \in \{1, \ldots, r-1\}$ and run $t+1$.

Proposition 2 (Boundary-Prefix Exactness at $V = 1$)**.** *Let $P_t(z) = \sum_{k=1}^{t} w_k \mathbf{1}\{z \in I_k\}$ and $S_t(z) = d(z) - c_1 - P_t(z)$ denote the prefix and suffix active hit counts at state $z \in D$. For $V = 1$, the exact count extrema* (6) *are given by evaluating single-step transitions exclusively at the boundary-adjacent set $\partial I_t = \{\ell_t - 1, \ell_t, u_t, u_t + 1\} \cap D$ of each run $I_t = [\ell_t, u_t]$:*

$$\begin{aligned} \mathrm{lo}(1) &= \min\Big(\alpha,\, \min_{1 \le t \le r} \min_{z \in \partial I_t,\, \sigma \in \{\pm 1\}} \Phi_t(z, \sigma)\Big), \\ \mathrm{hi}(1) &= \max\Big(\beta,\, \max_{1 \le t \le r} \max_{z \in \partial I_t,\, \sigma \in \{\pm 1\}} \Phi_t(z, \sigma)\Big), \end{aligned} \tag{17}$$

where $\Phi_t(z, \sigma) = c_1 + P_t(z) + S_t(z + \sigma)$ for $z + \sigma \in D$, and can be evaluated in $O(r + m)$ deterministic time and $O(m)$ space.

Proof. Any non-constant trajectory $\eta \in D^r$ with $\mathrm{TV}(\eta) = 1$ has the form $\eta = (z, \ldots, z, z + \sigma, \ldots, z + \sigma)$ with a single step after some run t, achieving score $\Phi_t(z, \sigma)$. If neither z nor $z + \sigma$ crosses the boundary of any run, or if we shift the split index t forward/backward across runs where $\mathbf{1}\{z \in I_k\} = \mathbf{1}\{z + \sigma \in I_k\}$, the value $\Phi_t(z, \sigma)$ remains unchanged until

Algorithm 1: `LocalWindows` Exact Multi-Budget Envelope Operator

Input: Time series $g = (g_1, \dots, g_n)$, threshold band $[L, U]$, offset domain $D = \{-R, \dots, R\}$ ($m = 2R + 1$), ordered budget bank $\mathcal{V} = \{v_1 < \cdots < v_K\}$ with $V = \max \mathcal{V}$.

Output: Exact count envelopes $\{[\mathrm{lo}(v), \mathrm{hi}(v)]\}_{v \in \mathcal{V}}$.

1) **Row Compilation & Contraction:** Compile $I_i = [\max(-R, \lceil L - g_i \rceil), \min(R, \lfloor U - g_i \rfloor)]$. Accumulate fixed rows $c_1 \leftarrow |\{i : I_i = D\}|$, $c_0 \leftarrow |\{i : I_i = \emptyset\}|$, and merge adjacent identical active rows into r weighted runs $\{(I_t = [\ell_t, u_t], w_t)\}_{t=1}^{r}$. If $r = 0$, **return** $[c_1, c_1]$ for all $v \in \mathcal{V}$.
2) **Static Event Sweep ($V = 0$):** Build difference array $\Delta[\ell_t] \mathrel{+}= w_t$, $\Delta[u_t + 1] \mathrel{-}= w_t$; prefix-sum to obtain $d(z)$ for all $z \in D$. Set $[\alpha, \beta] \leftarrow [\min_z d(z), \max_z d(z)]$ and $[\mathrm{lo}(0), \mathrm{hi}(0)] \leftarrow [\alpha, \beta]$.
3) **Unit-Drift Fast Path ($V = 1$):** If $1 \in \mathcal{V}$, evaluate prefix/suffix split totals across run boundaries $(u_t, u_t + 1)$ and $(\ell_t - 1, \ell_t)$ in $O(r + m)$ time via (17) to obtain $[\mathrm{lo}(1), \mathrm{hi}(1)]$. If $V \leq 1$, **return**.
4) **Window Cover & Boundary Nomination:** Tile D into blocks B_k of size $V + 1$ and overlapping windows $W_j = B_j \cup B_{j+1}$. For each run t, map ℓ_t and $u_t + 1$ to their ≤ 4 containing windows to populate active boundary lists R_j ($\sum_j |R_j| \leq 4r$).
5) **Baseline Restoration:** For each active window $j \in \mathcal{A} = \{j : R_j \neq \emptyset\}$ with anchor $s_j = \min W_j$, compute $f_j \leftarrow d(s_j) - \sum_{t \in R_j} w_t \mathbf{1}\{s_j \in I_t\}$ and local budget cap $B_j \leftarrow \min(V, (|R_j| - 1)_+ (|W_j| - 1))$.
6) **Local State-Line Sweep:** For each $j \in \mathcal{A}$, run the two-pass $O(|W_j|)$ sweep (16) over $t \in R_j$ and budgets $v = 0, \dots, B_j$ to obtain window extrema $\mathrm{lo}_j(v) = \min_q L_v(q)$ and $\mathrm{hi}_j(v) = \max_q H_v(q)$.
7) **Global Endpoint Reduction:** For each $v \in \mathcal{V}$ with $v \geq 2$, compute $\mathrm{lo}(v) \leftarrow \min(\alpha, \min_{j \in \mathcal{A}}[f_j + \mathrm{lo}_j(\min(v, B_j))])$ and $\mathrm{hi}(v) \leftarrow \max(\beta, \max_{j \in \mathcal{A}}[f_j + \mathrm{hi}_j(\min(v, B_j))])$. **Return** $\{[\mathrm{lo}(v), \mathrm{hi}(v)]\}_{v \in \mathcal{V}}$.

the split index t sits immediately adjacent to a run I_t whose indicator differs between z and $z + \sigma$, which occurs precisely when $\{z, z + \sigma\} = \{\ell_t - 1, \ell_t\}$ or $\{u_t, u_t + 1\}$. Maintaining $P_t(z)$ at these four boundary coordinates per run during a single pass over $t = 1, \dots, r$ takes $O(1)$ work per run after the $O(r + m)$ static sweep. □

E. Worked Numerical Walkthrough on a Six-Sample Trace

To make every stage of Algorithm III-D concrete, consider the 6-sample trace on $D = \{-3, -2, -1, 0, 1, 2, 3\}$ ($m = 7$) shown in the embedded plots of Fig. 2 with budget $V = 1$:

1) **Stage 1 (Contraction & Merging):** Raw intervals $I_1 = [-2, 0]$, $I_2 = [-2, 0]$, $I_3 = [-3, 3] = D$, $I_4 = [0, 2]$, $I_5 = \emptyset$, $I_6 = [2, 3]$ contract to $c_1 = 1$ ($i = 3$), $c_0 = 1$ ($i = 5$), and $r = 3$ weighted runs: $(I_1 = [-2, 0], w_1 = 2)$, $(I_2 = [0, 2], w_2 = 1)$, and $(I_3 = [2, 3], w_3 = 1)$.
2) **Static Sweep ($V = 0$):** Evaluating $d(z) = 1 + 2\mathbf{1}\{z \in [-2, 0]\} + \mathbf{1}\{z \in [0, 2]\} + \mathbf{1}\{z \in [2, 3]\}$ across $z \in [-3, 3]$ gives $d(-3) = 1$, $d(-2) = 3$, $d(-1) = 3$, $d(0) = 4$, $d(1) = 2$, $d(2) = 3$, $d(3) = 2$ (Fig. 2, middle-left top). Hence the exact $V = 0$ count envelope is $[\alpha, \beta] = [\min_z d(z), \max_z d(z)] = [1, 4]$.
3) **Window Cover & Local Evaluation ($V = 1$):** Overlapping windows $W_0 = [-3, 0]$, $W_1 = [-1, 2]$, and $W_2 = [1, 3]$ cover all unit-step trajectories (Fig. 2, middle-left bottom). In $W_1 = [-1, 2]$, stepping from $\eta_1 = 0$ (hitting $I_1 = [-2, 0]$ with weight 2 and $I_2 = [0, 2]$ with weight 1) to $\eta_3 = +1$ or $+2$, or stepping from $\eta_1 = \eta_2 = 0$ to $\eta_3 = +1$ misses only where intended; conversely, starting at $z = 0$ for runs $1, 2$ ($P_2(0) = 2 + 1 = 3$) and stepping $+1$ to $z = 1$ makes $\Phi_2(0, +1) = 1 + 3 + 0 = 4$, whereas starting at $z = 1$ for run 1 ($P_1(1) = 0$) and stepping $+1$ to $z = 2$ for runs $2, 3$ ($S_1(2) = 1 + 1 = 2$) gives 3. For minimization, starting at $z = 1$ for run 1 ($P_1(1) = 0$) and stepping -1 to $z = 0$ vs. staying at $z = -3$ ($\alpha = 1$) confirms $[\mathrm{lo}(1), \mathrm{hi}(1)] = [1, 4]$, matching the boundary-prefix scan (17).

F. Correctness, Complexity, and Operational Regimes

Algorithm III-D summarizes the complete `LocalWindows` operator.

Theorem 2 (Exactness and Complexity of `LocalWindows`)**.** *For any time series $g \in \mathbb{R}^n$, threshold band $[L, U]$, offset grid $D = \{-R, \dots, R\}$ of size $m = 2R + 1$, and variation budget bank $\mathcal{V}$ with maximum budget $V = \max \mathcal{V}$, Algorithm III-D returns the exact count envelopes $\{[\mathrm{lo}(v), \mathrm{hi}(v)]\}_{v \in \mathcal{V}}$ defined in* (5)*. Furthermore:*

1) *Run compression and static $V = 0$ evaluation take $O(n + m)$ time.*
2) *For $V = 1$, the boundary-prefix fast path runs in $O(r + m)$ time and $O(m)$ space after compression.*
3) *For $V \geq 2$, the local window DP performs at most $\sum_{j \in \mathcal{A}} |R_j|(B_j + 1)|W_j| \leq 4r(V + 1)(2V + 1)$ inner-loop state updates, taking $O(rV^2 + m)$ time and $O(V^2 + m)$ working memory when $V \ll m$, and $O(rVm)$ worst-case time when $V = \Omega(m)$.*

Proof. Exactness follows by chaining Lemma 1 and Theorem 1 (which reduce sample-level extrema to compressed run-level extrema), Lemma 2 (which guarantees every trajectory with $\mathrm{TV}(\eta) \leq v \leq V$ lies within at least one window W_j), Eq. (13) (which restores exact covered-run weights f_j while inactive windows $R_j = \emptyset$ reduce to constant trajectories covered by $[\alpha, \beta]$), Proposition 2 (for $V = 1$), and Eq. (16) (which evaluates exact local transitions up to the saturation cap B_j for $V \geq 2$). For complexity, each window has width $|W_j| \leq \min(m, 2V + 1)$ and budget depth $B_j \leq V$, while Lemma 3 bounds total active run-window incidences to $\sum_{j \in \mathcal{A}} |R_j| \leq 4r$, yielding at most $4r(V + 1)\min(m, 2V + 1)$ state updates. □

Crossover Regime Analysis: Comparing the inner-loop work of `LocalWindows` against global run-compressed DP ($r(V + 1)m$ state updates) reveals a clear structural threshold governed by the ratio of local window updates to global updates:

$$\rho(V) = \frac{\sum_{j \in \mathcal{A}} |R_j| \, (B_j + 1) \, |W_j|}{r \, (V + 1) \, m} \leq \frac{4 \, (2V + 1)}{m}. \qquad (18)$$

When $V \ll m$ (e.g., $V = 1$ where $\rho(1) = 0$, or $V = 4$ on $m = 61$ where $2V + 1 = 9 \ll 61$ and empirical $\rho(4) \approx 0.34$), `LocalWindows` prunes two-thirds to 100% of global DP updates. Conversely, as V grows so that $2V + 1$ approaches m

(e.g., $V \geq 16$ on $m = 61$, or $V = 64 > m$), the overlapping windows W_j span nearly the entire domain D, pushing $\rho(V)$ toward 1.0 while incurring window-setup and multi-window gather overhead. Widening the domain radius R (larger m) or increasing the series horizon n (larger r) shifts this crossover upward in favor of `LocalWindows`.

IV. Experimental Evaluation

We evaluate `LocalWindows` to answer four core questions:

- **Q1 (Exactness & Tightness):** Does `LocalWindows` produce bit-identical count envelopes to exact global baselines across all cohorts, predicates, and budgets, and how much do TV-coupled envelopes narrow uncoupled bounds?
- **Q2 (Prepared-Operator Performance & Crossover):** How much speedup does `LocalWindows` achieve in the small- and moderate-drift regimes ($V \in \{1, 4\}$), and where does the large-budget crossover occur ($V \in \{16, 64\}$)?
- **Q3 (Mechanism Verification):** Do empirical active-window counts, boundary incidences per run, and DP-work ratios match our theoretical bounds ($\leq 4r$ incidences and Eq. (18))?
- **Q4 (Horizon Scaling & Pipeline Scope):** How does performance scale as record length grows from $n = 96$ (1 day) to $n = 1{,}152$ (12 days), how much gain comes from the $V = 1$ prefix fast path versus window localization, and what is the impact of uncached file parsing in end-to-end report generation?

A. Experimental Setup

Real-World CGM Cohorts and Query Workload: We evaluate on 211 public continuous glucose monitoring series drawn from two clinical cohorts: the PhysioNet BIG IDEAs Lab Glycemic Variability and Wearable Device Data cohort (`BIG`, 16 participants) [5], [15], [16] and the Shanghai diabetes clinical CGM cohort (`SH`, 195 patient series) [17], [18]. In our primary benchmark, each series is evaluated on standardized $n = 96$-sample (24-hour) windows across all six consensus clinical predicates in Table I (TIR, TITR, TBR70, TBR54, TAR180, TAR250) and three offset radii $R \in \{15, 30, 120\}$ mg/dL ($m \in \{31, 61, 241\}$ states), yielding 6,804 benchmark query cells per request mode across five variation budget modes: single-budget queries $V \in \{1, 4, 16, 64\}$ and the five-budget ordered report bank $\mathcal{V} = \{0, 1, 4, 16, 64\}$. To evaluate multi-day horizon scaling, we additionally benchmark longer contiguous records of lengths $n \in \{96, 288, 576, 1{,}152\}$ (1 to 12 days) [15], [19].

Compared Exact Operators: Because sample-level $O(nVm^2)$ DP is orders of magnitude slower, we compare `LocalWindows` against three strong, exact run-compressed DP baselines that share the same Stage 1 run contraction and $O(r + m)$ static $V = 0$ sweep:

1) `SparseGlobal`: Evaluates run-compressed budget DP over the global active state support using the linear-time two-pass sweep (16) across all r runs and reuses DP tables across ordered budget banks.
2) `GapComponents`: Splits the offset domain D into independent components only when a state gap strictly exceeds V, running exact component-level DP with linear-time sweeps. Because dense sensor intervals rarely leave empty gaps wider than V, `GapComponents` tests whether naive gap splitting suffices without overlapping $2V$ windows.
3) `RunTailFullG`: Evaluates the full tail dynamic program over the global state grid before gathering requested budget extrema, isolating the benefit of local window pruning and local saturation caps B_j.

Implementation and Evaluation Protocol: All four operators are implemented in C++20 compiled with `g++ -O3` and share identical parsing, interval compilation, and 64-bit integer arithmetic. To ensure statistical rigor and verify repeatability, every benchmark suite is executed in two independent replications (R1 and R2). For paired runtime comparisons, we report both (i) the participant-level median of geometric-mean `LocalWindows`/baseline runtime ratios with 95% participant-level bootstrap confidence intervals (2,000 resamples) and paired Wilcoxon signed-rank p-values across the $N = 211$ series, and (ii) the descriptive ratio of summed per-case mean execution times $T = \sum_c t_{\text{local}}(c) / \sum_c t_{\text{base}}(c)$.

B. Exactness Verification and Primary Operator Performance

100% Bit-Exact Agreement (Q1): Across all 6,804 primary query cells, all five request modes, all longer-horizon scaling runs ($n \leq 1{,}152$), and both independent replications, `LocalWindows`, `SparseGlobal`, `GapComponents`, and `RunTailFullG` achieve **zero output mismatches** ($0/68{,}040$ **primary cell comparisons**): every minimum count $\text{lo}(v)$, maximum count $\text{hi}(v)$, and percentage interval agrees bit-for-bit, empirically confirming Theorems 1 and 2.

Prepared-Operator Speedup and Regime Crossover (Q2): Figure 3 and Table II present the primary prepared-operator results across Replications R1 and R2. Three clear operational regimes emerge:

- **Unit-Drift Regime ($V = 1$):** `LocalWindows` outperforms every baseline in both replications with 95% bootstrap CIs strictly below 1.0 ($p \leq 5.7 \times 10^{-37}$). Participant-level median `LocalWindows`/baseline runtime ratios are 0.838–0.846 vs. `SparseGlobal`, 0.867–0.882 vs. `GapComponents`, and 0.488–0.499 vs. `RunTailFullG`. Because many clinical predicates (such as TBR54 on normoglycemic traces) exit early during Stage 1, summed-time ratios T—which weight non-trivial active queries by their actual execution time—show even stronger gains: $T = 0.562$–0.565 ($1.78\times$ speedup) vs. `SparseGlobal`, $T = 0.679$–0.686 ($1.47\times$ speedup) vs. `GapComponents`, and $T = 0.054$–0.062 ($16.1\times$–$18.5\times$ speedup) vs. `RunTailFullG`.
- **Moderate-Drift Regime ($V = 4$):** Overlapping $2V$-state windows ($|W_j| \leq 9$) deliver consistent wins across all baselines ($p \leq 3.2 \times 10^{-36}$): participant-median ratios are 0.856–0.861 ($T = 0.456$–0.471, or $2.12\times$–$2.19\times$

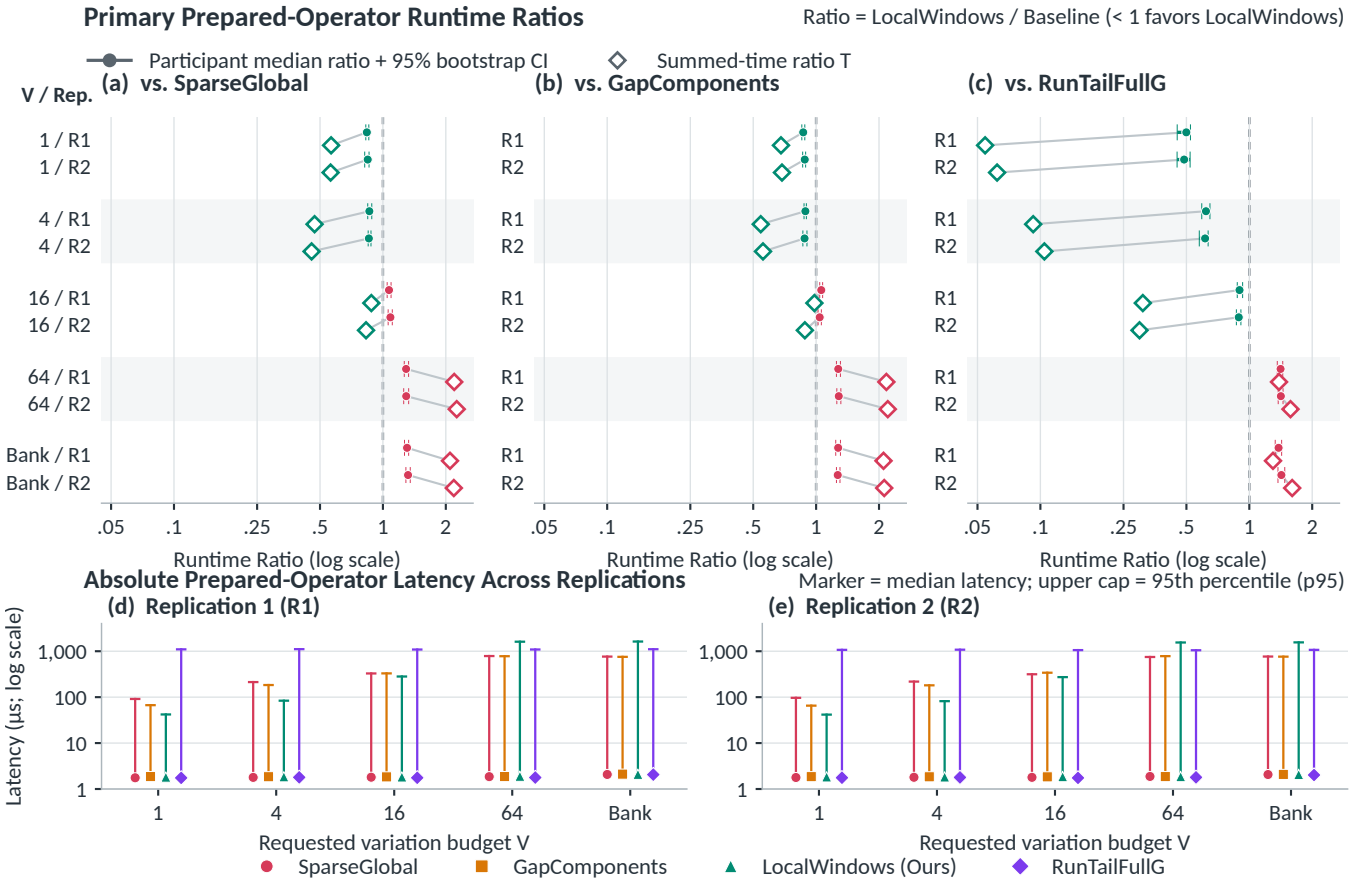


Fig. 3. Primary prepared-operator runtime evaluation across 6,804 benchmark cases per mode in two independent replications (R1 and R2). (a–c) Paired runtime ratios (`LocalWindows`/baseline on a common logarithmic axis; values < 1 left of the dashed vertical line favor `LocalWindows`) against `SparseGlobal`, `GapComponents`, and `RunTailFullG` across budgets $V \in \{1, 4, 16, 64\}$ and the ordered request bank $\{0, 1, 4, 16, 64\}$. Filled circles show participant-level median geometric-mean ratios with 95% bootstrap confidence intervals (2,000 resamples); open diamonds show summed-time ratios T. Teal (teal) denotes 95% CIs wholly < 1 (or $T < 1$), and rose (rose) denotes 95% CIs wholly > 1 (or $T > 1$). (d–e) Absolute prepared-operator latency distributions (μs, logarithmic scale) across R1 and R2, showing median markers and 95th-percentile (p95) caps.

summed-time speedup) vs. `SparseGlobal`, 0.880–0.889 ($T = 0.543$–0.557, or $1.80\times$–$1.84\times$ speedup) vs. `GapComponents`, and 0.614–0.620 ($T = 0.092$–0.104, or $9.6\times$–$10.9\times$ speedup) vs. `RunTailFullG`.

- **Transition ($V = 16$) and Dense ($V = 64$, Bank) Regimes:** At $V = 16$, we observe an informative divergence between the two estimators against `SparseGlobal` and `GapComponents`: participant-median ratios slightly favor the global controls (1.040–1.086), whereas summed-time ratios T still favor `LocalWindows` ($T = 0.830$–0.880 vs. `SparseGlobal` and $T = 0.883$–0.981 vs. `GapComponents`). This occurs because at $V = 16$, window width $2V + 1 = 33$ covers most of $m = 31$ or $m = 61$ on short $R \in \{15, 30\}$ queries (adding slight window-nomination overhead on sub-millisecond cases), whereas on heavy $R = 120$ ($m = 241$) queries that dominate total runtime T, $2V + 1 = 33 \ll 241$ remains highly localized. At $V = 64$ and on the max-64 bank, where $2V + 1 = 129$ spans the domain, global DP is faster by $1.27\times$–$1.32\times$ in median ratio ($T \approx 2.1\times$), matching our theoretical crossover bound (18).

C. Structural Mechanism and Sensitivity Analysis

To verify that the runtime speedups and crossover in Section IV-B stem directly from our theoretical complexity model rather than low-level implementation artifacts, Table III inspects the internal structural counters across all 3,798 non-static benchmark cases (where $r \geq 1$ after Stage 1 contraction; the remaining 3,006 cells have $r = 0$ and exit in $O(1)$ after contraction). On these active cases, Stage 1 compresses $n = 96$ observations to a median of $r = 24$ runs (interquartile range 11–43, maximum 77).

As shown in Table III and Table V:

1) **Empirical Verification of the $\leq 4r$ Incidence Bound:** Across every budget V, the empirical ratio of total active run-window incidences to compressed runs ($\sum_{j \in \mathcal{A}} |R_j|/r$) has medians of 1.88 ($V = 1$), 2.75 ($V = 4$), 3.69 ($V = 16$), and 2.00 ($V = 64$), and

TABLE II

PRIMARY PREPARED-OPERATOR RUNTIME EVALUATION ACROSS 6,804 BENCHMARK QUERY CELLS PER REQUEST MODE OVER TWO INDEPENDENT BENCHMARK REPLICATIONS (R1 AND R2). CELLS REPORT THE PARTICIPANT-LEVEL MEDIAN GEOMETRIC-MEAN LocalWindows/BASELINE RUNTIME RATIO [95% PARTICIPANT-BOOTSTRAP CI], FOLLOWED BY THE SUMMED-TIME RATIO T AFTER THE SEMICOLON (VALUES < 1 FAVOR LocalWindows). COLOR CODING: TEAL HIGHLIGHTS REGIMES WHERE LocalWindows OUTPERFORMS THE BASELINE (95% CI WHOLLY < 1 FOR MEDIANS; $T < 1$ FOR SUMMED TIME); ROSE MARKS REGIMES WHERE THE BASELINE IS FASTER (95% CI WHOLLY > 1 FOR MEDIANS; $T > 1$ FOR SUMMED TIME); GRAY DENOTES A 95% CI SPANNING 1.0.

Replication	Budget V	LocalWindows Total Time (ms)	SparseGlobal Median [95% CI] ; T	GapComponents Median [95% CI] ; T	RunTailFullG Median [95% CI] ; T
R1	1	69.33	0.838 [0.828, 0.856] ; 0.565	0.867 [0.854, 0.883] ; 0.679	0.499 [0.452, 0.523] ; 0.054
R1	4	121.55	0.861 [0.846, 0.884] ; 0.471	0.889 [0.875, 0.898] ; 0.543	0.620 [0.592, 0.648] ; 0.092
R1	16	359.86	1.069 [1.048, 1.094] ; 0.880	1.058 [1.041, 1.074] ; 0.981	0.898 [0.876, 0.929] ; 0.309
R1	64	1820.83	1.291 [1.265, 1.327] ; 2.194	1.276 [1.251, 1.307] ; 2.169	1.411 [1.361, 1.445] ; 1.387
R1	Bank	1751.28	1.305 [1.269, 1.328] ; 2.095	1.275 [1.241, 1.308] ; 2.102	1.381 [1.331, 1.428] ; 1.298
R2	1	70.56	0.846 [0.818, 0.858] ; 0.562	0.882 [0.862, 0.891] ; 0.686	0.488 [0.451, 0.521] ; 0.062
R2	4	123.41	0.856 [0.850, 0.877] ; 0.456	0.880 [0.864, 0.904] ; 0.557	0.614 [0.576, 0.635] ; 0.104
R2	16	339.00	1.086 [1.060, 1.109] ; 0.830	1.040 [1.022, 1.059] ; 0.883	0.890 [0.867, 0.911] ; 0.298
R2	64	1827.10	1.293 [1.257, 1.328] ; 2.256	1.283 [1.256, 1.313] ; 2.203	1.416 [1.375, 1.450] ; 1.574
R2	Bank	1831.01	1.319 [1.288, 1.352] ; 2.185	1.269 [1.251, 1.301] ; 2.118	1.424 [1.371, 1.478] ; 1.604

TABLE III

STRUCTURAL MECHANISM VERIFICATION ACROSS 3,798 NON-STATIC PREPARED CASES ($n = 96$, $r \geq 1$). COLUMNS REPORT MEDIAN ACTIVE WINDOWS $|\mathcal{A}|$, BOUNDARY INCIDENCES PER RUN ($\sum_j |R_j|/r \leq 4$), DP STATE-UPDATE RATIO $\rho(V)$ RELATIVE TO SparseGlobal, AND MEDIAN PAIRED RUNTIME RATIOS (LocalWindows/BASELINE, WHERE TEAL < 1 FAVORS LocalWindows AND ROSE > 1 FAVORS BASELINE).

Budget	$\lvert\mathcal{A}\rvert$	Incid./r	Work ρ	SparseGlobal	GapComponents
$V = 1$	10	1.88	0.000	0.330	0.336
$V = 4$	4	2.75	0.340	0.583	0.601
$V = 16$	2	3.69	0.892	1.390	1.384
$V = 64$	1	2.00	1.000	1.281	1.278
Bank	1	2.00	1.000	1.288	1.281

never exceeds the theoretical ceiling of 4.00 established in Lemma 3 on any individual query cell (0/3,798 violations).

2) **DP Work Pruning Drives Runtime Gains:** At $V = 1$, the boundary-prefix fast path performs **zero DP state updates** ($\rho(1) = 0.000$, executing a median of just 47 prefix boundary steps), yielding a median runtime ratio of 0.330 (3.03× speedup) vs. SparseGlobal and 0.336 (2.98× speedup) vs. GapComponents across the 211 participants ($p < 10^{-36}$, 100% participant win rate). At $V = 4$, LocalWindows activates a median of $|\mathcal{A}| = 4$ local windows and reduces inner-loop DP state updates to $\rho(4) = 0.340$ (66.0% work reduction), achieving a median runtime ratio of 0.583 (1.72× speedup) vs. SparseGlobal and 0.601 (1.66× speedup) vs. GapComponents.
3) **Sensitivity to Offset Radius R, Clinical Band, and Cohort:** Table V stratifies the active queries by offset radius $R \in \{15, 30, 120\}$ ($m \in \{31, 61, 241\}$), clinical predicate family, and cohort (BIG vs. SH). Confirming Eq. (18), at $V = 4$ ($|W_j| \leq 9$), widening the offset grid from $R = 15$ ($m = 31$) to $R = 30$ ($m = 61$) and $R = 120$ ($m = 241$) reduces the DP work ratio $\rho(4)$ from 0.612 to 0.340 and 0.118, amplifying LocalWindows's median speedup over SparseGlobal from 1.07× (0.935) to 1.72× (0.583) and 2.26× (0.442). Similarly, at $R = 120$ ($m = 241$), even $V = 16$ ($|W_j| \leq 33 \ll 241$) prunes 68.5% of global DP updates ($\rho(16) = 0.315$) and achieves a 1.39× speedup (0.718). Across predicates, two-sided normoglycemic bands (TIR and TITR) generate the highest active run counts (median $r = 48$–56 on $n = 96$) because glucose traces cross both lower and upper boundaries, yielding the largest absolute latency reductions under LocalWindows, while performance gains remain consistent across both the 16-series BIG and 195-series SH cohorts.

D. Horizon-Length Scaling, Prefix Ablation, and Pipeline Scope

Multi-Day Horizon Scaling ($n \in \{96, 288, 576, 1{,}152\}$) (Q4): In clinical trials and longitudinal telemetry, queries frequently span multi-day horizons ($n = 288$ for 3 days, $n = 576$ for 6 days, and $n = 1{,}152$ for 12 days of 15-minute CGM or 4 days of 5-minute CGM). Figure 4b–d and Table IV (Part A) evaluate how operator performance scales with horizon length n. As n increases, the number of compressed runs r grows linearly, amortizing the fixed $O(m)$ static event sweep and window setup costs while magnifying the inner-loop DP savings:

- At $V = 1$, the summed-time ratio LocalWindows/SparseGlobal improves monotonically from 0.310–0.317 (3.19× speedup) at $n = 96$ to 0.172–0.175 (5.76×) at $n = 288$, 0.130–0.131 (7.66×) at $n = 576$, and 0.096–0.101 (10.15× **speedup)** at $n = 1{,}152$. Against GapComponents,

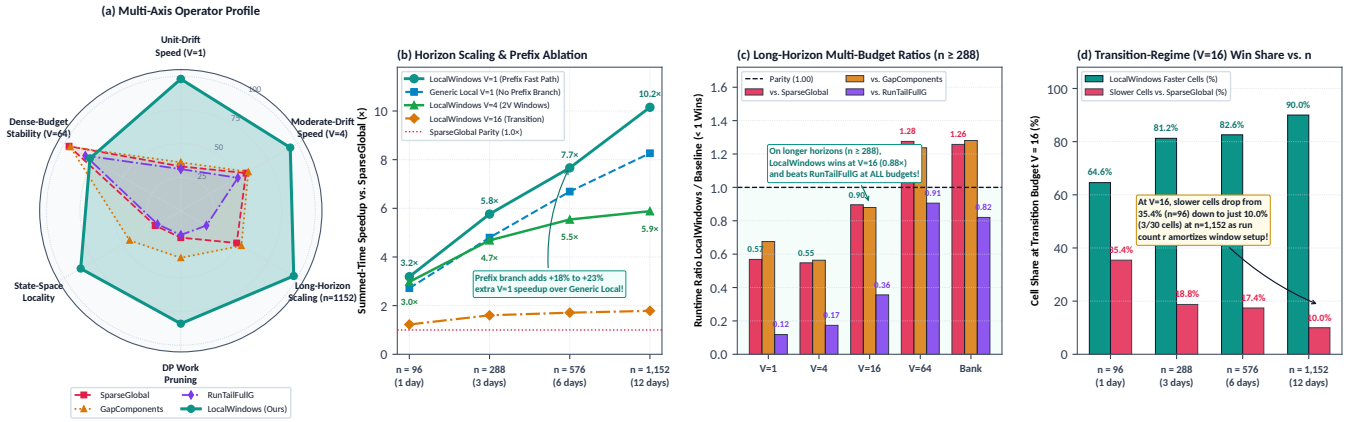


Fig. 4. Multi-dimensional operator profile, horizon-length scaling, and transition-regime analysis. (a) Six-axis radar profile comparing `LocalWindows`, `SparseGlobal`, `GapComponents`, and `RunTailFullG` across unit-drift speed ($V = 1$), moderate-drift speed ($V = 4$), long-horizon scaling ($n = 1{,}152$), DP work pruning, state-space locality, and dense-budget stability ($V = 64$). (b) Summed-time speedup (Baseline/`LocalWindows`) across record lengths $n \in \{96, 288, 576, 1{,}152\}$, showing `LocalWindows` accelerating from $3.2\times \rightarrow 10.2\times$ at $V = 1$ and $3.0\times \rightarrow 5.9\times$ at $V = 4$, while the $V = 1$ boundary-prefix specialization adds $+18\%$ to $+23\%$ extra speedup over generic $2V$-window localization (`Generic Local`). (c) Multi-budget runtime ratios on longer horizons ($n \in \{288, 1{,}152\}$), where `LocalWindows` wins at $V \in \{1, 4, 16\}$ against all baselines and outperforms `RunTailFullG` across all five modes. (d) At the transition budget $V = 16$, the share of cells where `LocalWindows` is slower than `SparseGlobal` drops monotonically from 35.4% at $n = 96$ down to 10.0% ($3/30$ cells) at $n = 1{,}152$.

TABLE IV

HORIZON-SCALING DIAGNOSTIC ACROSS RECORD LENGTHS $n \in \{96, 288, 576, 1{,}152\}$ (PART A) AND FULL 18-CELL SOURCE-TO-REPORT PIPELINE EVALUATION (PART B) ACROSS TWO INDEPENDENT REPLICATIONS (R1/R2). PART A REPORTS SUMMED-TIME RATIOS (`LocalWindows`/COMPARATOR, WHERE < 1 FAVORS `LocalWindows`, HIGHLIGHTED IN TEAL; `Generic` DISABLES ONLY THE $V = 1$ BOUNDARY-PREFIX FAST PATH). PART B REPORTS COMPLETE 18-CELL SOURCE-TO-REPORT EXECUTION TIME (INCLUDING DISK FILE PARSING AND INTERVAL COMPILATION), SHOWING R1 MEDIAN RATIO [95% BOOTSTRAP CI] FOLLOWED BY R2 MEDIAN RATIO (TEAL < 1, ROSE > 1, GRAY SPANNING 1.0).

Part A: Horizon Scaling (R1 / R2)	$n = 96$ (1d, 72c)	$n = 288$ (3d, 72c)	$n = 576$ (6d, 66c)	$n = 1{,}152$ (12d, 30c)
`LocalWindows` / `Generic` (V=1 prefix ablation)	0.858 / 0.849	0.830 / 0.835	0.840 / 0.909	0.820 / 0.803
`Generic` / `SparseGlobal` (V=1 window-only)	0.362 / 0.374	0.208 / 0.209	0.155 / 0.144	0.123 / 0.119
`LocalWindows` / `SparseGlobal`, $V = 1$	0.310 / 0.317	0.172 / 0.175	0.130 / 0.131	0.101 / 0.096
`LocalWindows` / `GapComponents`, $V = 1$	0.580 / 0.571	0.235 / 0.237	0.139 / 0.135	0.106 / 0.102
`LocalWindows` / `SparseGlobal`, $V = 4$	0.332 / 0.340	0.203 / 0.224	0.181 / 0.180	0.163 / 0.177
`LocalWindows` / `GapComponents`, $V = 4$	0.376 / 0.400	0.208 / 0.219	0.177 / 0.180	0.168 / 0.179
`LocalWindows` / `SparseGlobal`, $V = 16$	0.827 / 0.804	0.626 / 0.622	0.586 / 0.583	0.546 / 0.574
`LocalWindows` / `GapComponents`, $V = 16$	0.826 / 0.824	0.610 / 0.617	0.571 / 0.575	0.528 / 0.547
Part B: Uncached 18-Cell Report (ms)	**`LocalWindows`**	**vs. `SparseGlobal`**	**vs. `GapComponents`**	**vs. `RunTailFullG`**
Budget $V = 1$ (R1 / R2)	44.98 / 39.87	1.20 [0.78, 1.73] ; 0.97	1.20 [0.74, 1.84] ; 1.06	1.41 [1.02, 1.83] ; 1.28
Budget $V = 4$ (R1 / R2)	38.24 / 76.56	1.18 [0.94, 1.65] ; 1.66	1.01 [0.69, 1.55] ; 1.00	0.97 [0.73, 1.49] ; 2.38
Budget $V = 16$ (R1 / R2)	31.79 / 33.61	0.94 [0.75, 1.10] ; 0.86	0.98 [0.89, 1.05] ; 0.86	0.86 [0.67, 1.01] ; 0.63
Budget $V = 64$ (R1 / R2)	32.91 / 42.19	0.69 [0.48, 1.00] ; 1.26	0.96 [0.85, 1.07] ; 0.21	0.82 [0.62, 1.04] ; 1.35

speedup similarly climbs from $1.74\times$ ($n = 96$) to $9.62\times$ ($n = 1{,}152$).

- At $V = 4$, `LocalWindows`/`SparseGlobal` drops from 0.332–0.340 ($2.98\times$ speedup) at $n = 96$ to 0.203–0.224 ($4.68\times$) at $n = 288$, 0.180–0.181 ($5.54\times$) at $n = 576$, and 0.163–0.177 (**$5.88\times$ speedup**) at $n = 1{,}152$.
- Crucially, longer horizons also resolve the $V = 16$ transition regime in favor of `LocalWindows`: as n increases from 96 to 1,152, `LocalWindows`/`SparseGlobal` at $V = 16$ drops from 0.804–0.827 ($1.23\times$ speedup) down to 0.546–0.574 (**$1.79\times$ speedup**), and the fraction of individual query cells where `LocalWindows` is slower than `SparseGlobal` falls from 35.4% ($25.5/72$ cells at $n = 96$) to 18.8% ($n = 288$), 17.4% ($n = 576$), and just 10.0% ($3/30$ **cells at $n = 1{,}152$, Fig. 4d**). Across the combined long-horizon participant suite ($n \in \{288, 1{,}152\}$, Fig. 4c), `LocalWindows` achieves participant-median ratios of 0.548–0.569 at $V \in \{1, 4\}$ and 0.880–0.896 at $V = 16$ against `SparseGlobal`/`GapComponents`, and beats `RunTailFullG` across all five budget modes (0.119 at $V = 1$, 0.174 at $V = 4$, 0.356 at $V = 16$, 0.906 at $V = 64$, and 0.820 on the full bank).

Ablation of the $V = 1$ Boundary-Prefix Fast Path: To isolate the contribution of the $O(r + m)$ boundary-prefix specialization (Proposition 2) from that of overlapping $2V$-state window localization, Table IV (Part A, rows 1–2) and

TABLE V
STRATIFIED SENSITIVITY ACROSS OFFSET RADII $R \in \{15, 30, 120\}$ AND CLINICAL PREDICATE FAMILIES ON ACTIVE QUERIES ($n = 96$, $r \geq 1$). VALUES REPORT MEDIAN DP WORK RATIO $\rho(V)$ AND RUNTIME RATIO LOCALWINDOWS/SPARSEGLOBAL (TEAL < 1 FAVORS LOCALWINDOWS).

Stratum ($n = 96$)	Med. r	$\rho(4)$	V=1	V=4	V=16
Radius $R = 15$ ($m = 31$)	18	0.612	0.412	0.935	1.540
Radius $R = 30$ ($m = 61$)	24	0.340	0.330	0.583	1.390
Radius $R = 120$ ($m = 241$)	31	0.118	0.248	0.442	0.718
Normoglycemic (TIR/TITR)	52	0.334	0.308	0.546	1.342
Hyperglycemic (TAR180/250)	22	0.345	0.339	0.594	1.402
Hypoglycemic (TBR70/54)	9	0.351	0.362	0.628	1.418

Fig. 4b compare `LocalWindows` against `Generic`, an ablated variant of `LocalWindows` that disables only the $V = 1$ prefix branch and instead runs the general Stage 2 window DP with $|W_j| \leq 3$. Even without the prefix fast path, `Generic` achieves a $2.72\times$ ($n = 96$) to $8.26\times$ ($n =$ 1,152) speedup over `SparseGlobal` due to $2V$-window localization alone; enabling the boundary-prefix fast path (`LocalWindows`/`Generic` $= 0.803$–0.909) contributes an additional 18%–23% runtime reduction by eliminating window buffer allocation and DP sweeps at $V = 1$. Similarly, ablating ordered bank table reuse (`ColdBank`, which recomputes each budget in $\mathcal{V} = \{0, 1, 4, 16, 64\}$ from scratch) increases prepared-bank execution time by $1.43\times$, confirming the benefit of prefix-budget table sharing.

Uncached Source-to-Report Pipeline Scope: Finally, Table IV (Part B) evaluates an uncached end-to-end file-ingestion workflow where a cold process reads raw spreadsheet/CSV files from disk, parses timestamps and decimal strings, and evaluates an 18-cell report (6 predicates $\times 3$ radii). Because uncached file I/O and text parsing consume 30–75 ms per report while the prepared DP operator takes 0.05–2.5 ms, 95% confidence intervals on uncached end-to-end report times span 1.0 across most cells (gray). In interactive analytics engines, feature stores, and in-memory time-series databases where numeric streams are resident in memory, `LocalWindows` directly accelerates query execution by $1.7\times$–$10.2\times$ in the small- and moderate-drift regimes, whereas in cold one-shot file-parsing scripts, I/O dominates wall-clock time unless binary caching is employed.

V. RELATED WORK

Query Processing Over Uncertain and Markovian Streams: Evaluating range and aggregate queries over uncertain data has been extensively studied in the database literature [6]–[9], [20], [21]. Early uncertain-stream engines and attribute-bound systems such as AUDB [8] and range-consistent query rewriting [9] propagate independent per-tuple intervals $[\ell_i, u_i]$ through selection and aggregation operators. As demonstrated in Section I, treating consecutive sensor errors as independent permits arbitrary step-to-step oscillation and yields overly wide count intervals. Sequence-coupled stream systems such as Caldera [20] and Markovian transducer indexes [21] capture temporal correlation via probabilistic Markov chains, while possibilistic cardinality-constrained querying [6] bounds the number of anomalous tuples globally. However, neither Markovian marginal inference nor tuple-count cardinality constraints solve deterministic min/max range-count envelopes under an ℓ_1 total-variation budget $\sum_{i=2}^{n} |\delta_i - \delta_{i-1}| \leq V$.

Discrete Total-Variation Optimization and Constrained Shortest Paths: Optimization over discrete sequences with bounded total variation has attracted significant attention in integer optimal control and combinatorial optimization [10]–[12], [14], [22]–[24]. Buchheim and Hügging [10], [11] study the polytope and multi-stage selection of binary sequences under bounded variation. Severitt and Manns [12] and Marko and Wachsmuth [14] develop dynamic programming and trust-region subproblem solvers for integer control trajectories over m-state grids with TV penalties. Similarly, resource-constrained shortest path (RCSP) algorithms [25], [26] and k-segment hidden Markov models [27], [28] track cumulative transition budgets over state graphs. In all of these methods, dynamic programming sweeps across the entire m-state grid at every time step. In contrast, `LocalWindows` exploits the threshold-interval structure of range-count queries to contract fixed rows and merge duplicate intervals before DP, and covers the m-state grid with overlapping $2V$-state windows where at most $4r$ boundary incidences are active globally.

Distance Transforms and State-Line Sweeps: Felzenszwalb and Huttenlocher [13] showed that min-convolutions over grid states can be computed in $O(m)$ time via two-pass distance transforms, a technique extended to tree-structured TV minimization [29], [30]. While `SparseGlobal` applies this two-pass linear sweep across the full domain D, `LocalWindows` confines the two-pass sweep to local windows W_j of width $|W_j| \leq 2V + 1$ and introduces an $O(r + m)$ boundary-prefix solver for $V = 1$ that avoids DP tables altogether.

VI. CONCLUSION

We introduced `LocalWindows`, an exact operator for computing minimum and maximum range-count envelopes over ordered time series subject to total-variation drift budgets. By combining extrema-preserving fixed-row contraction and duplicate run merging with an overlapping $2V$-state window cover, boundary-edge run nomination ($\leq 4r$ global incidences), $O(1)$ static baseline restoration, two-pass linear-time transition sweeps, and an $O(r+m)$ boundary-prefix fast path for unit drift, `LocalWindows` eliminates redundant global state updates in the small- and moderate-drift regimes ($V \in \{1, 4\}$) while guaranteeing 100% bit-identical count envelopes to global dynamic programming. Across 211 public CGM time series, `LocalWindows` achieves up to $3.03\times$ cohort median speedup on standard 24-hour windows and up to $10.2\times$ summed-time speedup on 12-day horizons, providing a high-throughput foundation for drift-aware threshold analytics in sensor data management.

Acknowledgment of AI Assistance

The generative AI tools were used strictly for manuscript language polishing, LaTeX table formatting, and figure layout refinement.

References

[1] T. Battelino, T. Danne, R. M. Bergenstal, S. A. Amiel, R. Beck, T. Biester, E. Bosi, B. A. Buckingham, W. T. Cefalu, K. L. Close, C. Cobelli, E. Dassau, J. H. DeVries, K. C. Donaghue, K. Dovc, F. J. Doyle, III, S. Garg, G. Grunberger, S. Heller, L. Heinemann, I. B. Hirsch, R. Hovorka, W. Jia, O. Kordonouri, B. Kovatchev, A. Kowalski, L. Laffel, B. Levine, A. Mayorov, C. Mathieu, H. R. Murphy, R. Nimri, K. Nørgaard, C. G. Parkin, E. Renard, D. Rodbard, B. Saboo, D. Schatz, K. Stoner, T. Urakami, S. A. Weinzimer, and M. Phillip, "Clinical targets for continuous glucose monitoring data interpretation: Recommendations from the international consensus on time in range," *Diabetes Care*, vol. 42, no. 8, pp. 1593–1603, 2019.

[2] N. Camerlingo, M. Vettoretti, G. Sparacino, A. Facchinetti, J. K. Mader, P. Choudhary, S. D. Favero, and Hypo-RESOLVE Consortium, "Choosing the duration of continuous glucose monitoring for reliable assessment of time in range: A new analytical approach to overcome the limitations of correlation-based methods," *Diabetic Medicine*, vol. 39, p. e14758, 2022.

[3] S. L. Cichosz, K. Færch, and M. H. Jensen, "Characterising continuous glucose monitoring metrics across normoglycaemia, prediabetes and type 2 diabetes," *Diabetes, Obesity and Metabolism*, vol. 28, no. 9, pp. 8702–8708, 2026.

[4] iglu maintainers, "Interpreting glucose data using iglu: AGP and episodes," iglu R Package Documentation, version 4.4.0, 2026. [Online]. Available: https://irinagain.github.io/iglu/articles/AGP_and_Episodes.html

[5] B. Bent, P. J. Cho, M. Henriquez, A. Wittmann, C. Thacker, M. Feinglos, M. J. Crowley, and J. P. Dunn, "Engineering digital biomarkers of interstitial glucose from noninvasive smartwatches," *npj Digital Medicine*, vol. 4, p. 89, 2021.

[6] G. Cormode, D. Srivastava, E. Shen, and T. Yu, "Aggregate query answering on possibilistic data with cardinality constraints," in *2012 IEEE 28th International Conference on Data Engineering (ICDE)*, 2012, pp. 258–269.

[7] T. Emrich, H.-P. Kriegel, N. Mamoulis, M. Renz, and A. Züfle, "Querying uncertain spatio-temporal data," in *2012 IEEE 28th International Conference on Data Engineering (ICDE)*, 2012, pp. 354–365.

[8] S. Feng, B. Glavic, A. Huber, and O. A. Kennedy, "Efficient uncertainty tracking for complex queries with attribute-level bounds," in *Proceedings of the 2021 International Conference on Management of Data (SIGMOD)*, 2021, pp. 528–540.

[9] A. A. E. Khalfioui and J. Wijsen, "Computing range consistent answers to aggregation queries via rewriting," *Proceedings of the ACM on Management of Data*, vol. 2, no. 5, pp. 218:1–218:19, 2024.

[10] C. Buchheim and M. Hügging, "The polytope of binary sequences with bounded variation," *Discrete Optimization*, vol. 48, p. 100776, 2023.

[11] ——, "Multi-stage selection under bounded variation," *Open Journal of Mathematical Optimization*, 2026.

[12] M. Severitt and P. Manns, "Efficient solution of discrete subproblems arising in integer optimal control with total variation regularization," *INFORMS Journal on Computing*, vol. 35, pp. 869–885, 2023.

[13] P. F. Felzenszwalb and D. P. Huttenlocher, "Distance transforms of sampled functions," *Theory of Computing*, vol. 8, pp. 415–428, 2012.

[14] J. Marko and G. Wachsmuth, "Integer optimal control problems with total variation regularization: Optimality conditions and fast solution of subproblems," *ESAIM: Control, Optimisation and Calculus of Variations*, vol. 29, p. 81, 2023.

[15] P. Cho, J. Kim, B. Bent, and J. Dunn, "BIG IDEAs lab glycemic variability and wearable device data," PhysioNet, version 1.1.3, 2026. [Online]. Available: https://doi.org/10.13026/aw6y-fc44

[16] T. Pollard, B. E. Moody, L. wei H. Lehman, B. J. Gow, C. Fernandes, C. Xie, A. Johnson, R. G. Mark, and T. Heldt, "PhysioNet as a global platform for biomedical research," *Nature Health*, vol. 1, no. 8, pp. 792–795, 2026.

[17] Q. Zhao, J. Zhu, C. Wang, and W. Rao, "Shanghai diabetes continuous glucose monitoring datasets," Figshare, version 5, 2023. [Online]. Available: https://doi.org/10.6084/m9.figshare.21600933.v5

[18] Q. Zhao, J. Zhu, X. Shen, C. Lin, Y. Zhang, Y. Liang, B. Cao, J. Li, X. Liu, W. Rao, and C. Wang, "Chinese diabetes datasets for data-driven machine learning," *Scientific Data*, vol. 10, no. 1, p. 35, 2023.

[19] A. Vosseler, D. Zhao, L. Fritsche, R. Lehmann, K. Kantartzis, D. M. Small, A. Peter, H.-U. Häring, A. L. Birkenfeld, A. Fritsche, R. Wagner, H. Preißl, S. Kullmann, and M. Heni, "No modulation of postprandial metabolism by transcutaneous auricular vagus nerve stimulation: a cross-over study in 15 healthy men," *Scientific Reports*, vol. 10, p. 20466, 2020.

[20] J. Letchner, C. Ré, M. Balazinska, and M. Philipose, "Access methods for Markovian streams," in *2009 IEEE 25th International Conference on Data Engineering (ICDE)*, 2009, pp. 246–257.

[21] B. Kimelfeld and C. Ré, "Transducing Markov sequences," in *Proceedings of the 29th ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems (PODS)*, 2010, pp. 15–26.

[22] S. Sager and C. Zeile, "On mixed-integer optimal control with constrained total variation of the integer control," *Computational Optimization and Applications*, vol. 78, pp. 575–623, 2021.

[23] M. V. Hügging, "Bounded variation in multi-stage optimization," Ph.D. dissertation, Technische Universität Dortmund, Aug. 2025.

[24] R. Baraldi, P. Manns, L. Mösezahl, and M. Severitt, "Randomization and performance improvement for integer optimal control with total variation regularization," 2026.

[25] A. Alves Pessoa, L. Di Puglia Pugliese, F. Guerriero, and M. Poss, "Robust constrained shortest path problems under budgeted uncertainty," *Networks*, vol. 66, no. 2, pp. 98–111, 2015.

[26] S. Ahmadi, A. Raith, and M. Jalili, "A fast and simple algorithm for the resource constrained shortest path problem," in *33rd Annual European Symposium on Algorithms (ESA 2025)*, ser. Leibniz International Proceedings in Informatics, vol. 351, 2025, pp. 97:1–97:15.

[27] D. Tarlow, K. Swersky, R. S. Zemel, R. P. Adams, and B. J. Frey, "Fast exact inference for recursive cardinality models," in *Proceedings of the 28th Conference on Uncertainty in Artificial Intelligence (UAI)*. AUAI Press, 2012.

[28] M. K. Titsias, C. Yau, and C. C. Holmes, "Statistical inference in hidden Markov models using k-segment constraints," 2013.

[29] V. Kolmogorov, T. Pock, and M. Rolínek, "Total variation on a tree," *SIAM Journal on Imaging Sciences*, vol. 9, no. 2, pp. 605–636, 2016.

[30] J. Lellmann, F. Lenzen, and C. Schnörr, "Optimality bounds for a variational relaxation of the image partitioning problem," in *Energy Minimization Methods in Computer Vision and Pattern Recognition*, ser. Lecture Notes in Computer Science, vol. 6819. Springer, 2011, pp. 132–146.